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Opportunistic behaviour behind corporate digitalization disclosure: The moderating role of economic policy uncertainty

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ABSTRACT

We identify the opportunistic behavior behind corporate digitalization disclosure (CDD) from an insider trading perspective and the moderating effect of economic policy uncertainty (EPU) on this relationship using a sample of listed enterprises in China from 2009 to 2022. We find that insider opportunistic trading behavior significantly explains the phenomenon of more CDD in the prior period management discussion and analysis (MD&A) of the annual report, driven by CDD can elevate investor sentiment. The moderating model shows that EPU worsens the observed phenomenon. Overall, our findings suggest insider may exploit digital trends for self-interest in emerging capital markets.

1. Introduction

Owing to the widespread application of digital technologies such as the Internet, big data, artificial intelligence, Internet of Things, cloud computing, and blockchain, the world has entered the digital economy era (Xia et al., 2023). Increasingly, companies are using digital technologies for innovation and disclosing digitalization information in their annual report's management discussion and analysis (MD&A) section. Most studies interpret firms' eagerness to disclose digital information as a positive signal to reduce information asymmetry and improve the corporate information environment in emerging capital markets (Wu et al., 2022; Niu et al., 2023; Cui and Wang, 2023). However, based on economic self-interest principle, insiders often use private idiosyncratic information for personal stock market gain (Cohen et al., 2012; Batten et al., 2021). Therefore, there is a dire need to determine the motives behind corporate digitalization disclosure (CDD) in MD&A for emerging capital markets with limited investor protection mechanisms.

Sawicki and Shrestha (2008) found that insiders manipulate financial disclosure to inflate the stock price before the reduction to maximize returns. Besides financial information, non-standardized textual information in MD&A of annual reports has become crucial for decision-making. Existing studies show that underperforming firms reduce their annual reports' readability by increasing text length (Li, 2008), and management accurately estimates the trade timings to maximize reduction in holdings to yield maximum returns by disseminating favorable information (Cheng and Lo, 2006; Brockman et al., 2008; Billings and Buslepp, 2016; Bushee et al., 2023). Additionally, because textual information is most often descriptive and subjective, regulators are unlikely to identify disclosure violations in emerging markets such as China. Thus, unlike financial information, MD&A's textual information adds substantial management discretion to report strategically and manipulate market expectations (Davis et al., 2012).

In the digital era, the terms "Digital Economy" and "Digital Transformation" have garnered media and investors' attention.

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Enterprises' digital transformation align with digital economy trends, leading investors to have positive expectations from these firms' development prospects. The deep integration of the digital and real economies complicates external investors' ability to accurately estimate the extent of enterprise digital transformation. Literature suggests that firms manipulate digitalization information to cater to investors (Zhou et al., 2024), especially when the CEO faces promotion pressure (Zhang et al., 2023). Therefore, the underlying motivation for CDD, whether public or private interests, warrants further exploration. This study examines the CDD motivations by testing the relationship between insiders opportunistic trading behavior and the digitalization information disclosure levels. If CDD aims to improve the firm's information environment, then insiders opportunistic trading should be independent of CDD levels, implying no significant correlation between them. Conversely, if CDD is motivated by insider self-interest, there is a significant positive relationship between the insiders' opportunistic trading behavior and the level of CDD, indicating that the CDD in MD&A was a deliberate pre-disclosure to response to capital market hotspots in line with later insiders' reduction.

In particular, economic policy uncertainty (EPU) can worsen information asymmetry between firms and external stakeholders, thereby complicating retail investors' understanding of the firms' actual operating situation and supervision of insiders. This expands insiders' information advantage (Nagar et al., 2019) and creates favorable conditions for insiders' self-interested behavior. Furthermore, higher EPU increases the likelihood of corporate executive change (Huang et al., 2021), and executives tend to gain excess returns as personal psychological compensation before leaving their jobs. Therefore, EPU strengthens the incentives of insiders, such as executives, to use CDD for opportunistic trading behavior, thus positively affecting the relationship between them.

This study's contributions include, first, revealing the dark side of CDD via insider trading, implying that "digital transformation" could be a tool for insider's conceptual speculation, paving the way for subsequent research. Second, earlier studies have examined the impact of information disclosure on insider trading, focusing on information readability, timing, and tone (Cheng and Lo, 2006; Brockman et al., 2008; Kraft et al., 2014; Billings and Buslepp, 2016). This study suggests that management might use "hot spots" to manipulate information to guide external investors toward a higher company stock valuation, thereby enriching existing studies on information disclosure and insider trading. Third, this finding offers useful insights for researchers of digitalization disclosure, external investors, and regulators in rational interpretation of corporate digitalization information, establish sound investment concepts, and enhance disclosure norms.

2. Research design

2.1. Data

This study uses the data of Chinese listed enterprises from 2009 to 2022, excluding financial firms, firms with ST, and those with missing relevant data to prevent the bias of regression results. The final sample consisted of 34,750 observations. Annual report data of listed firms is sourced from the Shanghai Stock Exchange and Shenzhen Stock Exchange (SZSE), with other financial data obtained from the China Stock Market & Accounting Research database. All continuous variables are winsorized at the 1 % and 99 % levels.

2.2. Variables

2.2.1. Explained variable: corporate digitalization disclosure

Following existing studies (Wu et al., 2022), we used the natural logarithm of one plus the frequency of words associated with "digital transformation" in MD&A of listed enterprises' annual reports as a proxy for the CDD.

2.2.2. Explanatory variable: insiders opportunistic trading

This study focuses on insiders including "major shareholders, directors, supervisors, executives and their relatives who hold more than 5 % of the shares." Opportunistic stock sales by insiders can yield higher excess returns, while the excess returns on normal share reductions are almost zero (Cohen et al., 2012). Based on earlier studies (Cohen et al., 2012; Huddart and Ke, 2007), the reduction announcement date is adopted as the base date. An opportunistic reduction is when insiders stock-selling anticipate negative stock

Table 1
Definitions of variables.

Variables	Definitions
CDD	The natural logarithm of one plus the frequency of digitalization keywords in MD&A.
Times	The natural logarithm of one plus the frequency of insider opportunistic stock-selling within 30 days after the annual report publication.
Amounts	The natural logarithm of one plus the amount of insider opportunistic stock-selling within 30 days after the annual report publication.
Size	The natural logarithm of total assets.
Lev	The total debt divided by total assets.
ROA	The net income divided by total assets.
TBQ	The ratio of market capitalization over total assets.
TOP1	The shareholding ratio of the largest shareholder.
IIP	The shareholding ratio of institutional investors.
Dual	The value is 1 if the chairman and general manager are one person, otherwise the value is 0.
MDP	The ratio of male members over boards of directors.
Big4	The value is 1 if the firm is audited by one of the four biggest audit firms, otherwise the value is 0.
Vol	The standard deviation of daily stock returns.

returns in the following month, otherwise it represents a normal reduction, and the frequency (*Times*) and the amount (*Amounts*) of reduction are estimated, respectively. Correspondingly, this study investigates insider trading occurring within 30 days after the publication of annual report. Besides, in the robustness tests, the trading window is extended to 60 and 90 days.

2.3. Empirical model

To examine the relation between CDD and insider trading, we follow Rogers (2008) to construct the following model:

$$CDD_{i,t-1} = \beta_0 + \beta_1 Times/Amounts_{i,t} + X_{i,t-1} + YearFE + IndustryFE + \varepsilon_{i,t-1} \quad (1)$$

where *i* and *t/t-1* are the firm and year, respectively. *CDD* is the explained variable. We used the natural logarithm of one plus the frequency of words associated with digitalization in MD&A as a proxy variable (Wu et al., 2022). The explanatory variables, *Times* and *Amounts*, represent the frequency and amount of insider opportunistic stock-selling in the 30 days after publication of the annual report. *X* represents a set of control variables (Zhao et al., 2021; Zhou et al., 2024). ε is the random error term. Table 1 shows the definitions of the variables.

3. Empirical results

3.1. Descriptive statistics

Table 2 shows the descriptive statistics for the main variables. Essentially, the mean values of *Times* and *Amounts* for sample firms are 0.028 and 0.423, respectively. Moreover, the mean value of *CDD* for sample enterprises is 0.910, with a standard deviation of 1.179, indicating that *CDD* has a large variability within the sample.

3.2. Main results

Table 3 shows the regression results. Noticeably, the coefficients of *Times* and *Amounts* are all significantly positive, with or without control variables. This implies that insider opportunistic trading plays an important role in explaining the phenomenon of increased CDD in prior period MD&A. In other words, insiders tend to disclose more information on digitalization strategically before reducing their holdings.

3.3. Robustness tests

3.3.1. Alternative measures of key variables

One is to use an alternative proxy for CDD. Drawing on earlier studies, we construct model (2) to measure the abnormal level of CDD (*Ab_CDD*), where *CDD_Ind* is the median *CDD* of the firms in the same industry-year, $DIGINV_{i,t}$ and $DIGINV_{i,t+1}$ measure the firm's digital investment amount to the ratio of total assets for the current and following years, *Year, Industry,* and *Province* are year-, industry-, and province-fixed effects, respectively. The residual from the regression of model (2) is *Ab_CDD* and we use *Ab_CDD* as an explained variable to run model (1), as shown in columns (1)–(2) of Table 4. The second alternative is to change the measurement of insider trading. We use the ratio of the frequency of insider opportunistic reductions to the total frequency of sales to measure *Times* and use the ratio of the number of insider opportunistic reductions to the total number of shares to measure *Amounts*. Columns (3)–(4) of Table 4 show the results.

$$CDD_{i,t} = \beta_0 + \beta_1 CDD_Ind_{i,t} + \beta_2 DIGINV_{i,t} + \beta_3 DIGINV_{i,t+1} + YearFE + IndustryFE + ProvinceFE + \varepsilon_{i,t} \quad (2)$$

Table 2
Descriptive statistics of the main variables.

Variable	N	Mean	SD	Min	Median	Max
CDD	34,750	0.910	1.179	0.000	0.000	4.394
Times	34,750	0.028	0.158	0.000	0.000	1.099
Amounts	34,750	0.423	2.363	0.000	0.000	15.287
Size	34,750	22.132	1.287	19.868	21.939	26.153
Lev	34,750	0.416	0.206	0.050	0.409	0.873
ROA	34,750	0.041	0.056	−0.203	0.040	0.195
TBQ	34,750	2.055	1.301	0.866	1.635	8.520
TOP1	34,750	0.346	0.149	0.087	0.325	0.748
IIP	34,750	0.453	0.253	0.003	0.468	0.935
Dual	34,750	0.288	0.453	0.000	0.000	1.000
MDP	34,750	0.811	0.112	0.500	0.823	1.000
Big4	34,750	0.059	0.236	0.000	0.000	1.000
Vol	34,750	0.034	0.023	0.013	0.029	0.194

Table 3
Baseline results.

	(1)	(2)	(3)	(4)
Variables	CDD	CDD	CDD	CDD
Times	0.159***		0.146***	
	(4.11)		(3.88)	
Amounts		0.010***		0.009***
		(3.98)		(3.68)
Size			0.111***	0.111***
			(8.70)	(8.69)
Lev			−0.258***	−0.258***
			(−3.90)	(−3.90)
ROA			−0.129	−0.129
			(−0.76)	(−0.76)
TBQ			0.036***	0.036***
			(4.49)	(4.49)
TOP1			−0.167**	−0.167**
			(−2.08)	(−2.08)
IIP			−0.110**	−0.110**
			(−2.01)	(−2.01)
Dual			0.114***	0.114***
			(5.03)	(5.03)
MDP			−0.514***	−0.514***
			(−5.24)	(−5.24)
Big4			−0.084**	−0.084**
			(−2.07)	(−2.07)
Vol			−0.137	−0.137
			(−0.52)	(−0.51)
Constant	0.905***	0.905***	−1.016***	−1.014***
	(78.10)	(78.06)	(−3.76)	(−3.76)
Year FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
N	34,750	34,750	34,750	34,750
Adj. R^2	0.355	0.355	0.365	0.365

Notes: Parentheses denote t-values adjusted for firm-level clustering; *, **, and *** stands for significant at the 10 %, 5 %, and 1 % levels, respectively. The following tables are the same.

3.3.2. *Extension of trading window*

We extend the trading window from 30 to 60 and 90 days after the annual report publication, as shown in columns (5)–(8) of Table 4, respectively.

3.3.3. *Changing the scope of insiders*

Given executives' ability to directly influence disclosure decisions and likely access to private information, we limit the scope of insiders to the executive level and use model (1) for the regression (columns (9)-(10), Table 4).

3.3.4. *Propensity score matching*

To mitigate selection bias, we use firms with opportunistic insider trading as a treatment group and other firms as the control group, select model (1) control variables as covariates to estimate propensity scores using the Logit model, and then match the treatment group to the same year-industry control group using the nearest-neighbor matching method with 1:1, 1:2, and 1:3, respectively, to generate a new regression sample, as shown in Panel A of Table 5.

3.3.5. *Adding more control variables and firm-fixed effects model*

To mitigate the missing variables issue, we control for the positive tone of the MD&A (*Tone*), the readability of the MD&A (*Readability*), and stock price volatility (*Spvol*). Specifically, *Tone* is measured using the ratio of the difference between the word frequency of positive words and negative words to the sum of the word frequency of positive words and negative words in the MD&A, *Readability* is measured using the frequency of word-pair collocation order in the MD&A text in the corpus, *Spvol* is measured using the standard deviation of the weekly closing price (columns (1)-(2) of Table 5, Panel B). Further, we control for firm-fixed effects in the regression model (columns (3)-(4) of Table 5, Panel B).

3.3.6. *Placebo test*

To account for the interference of other unobservable factors in the findings, we exclude this potential possibility through a placebo test randomly swaps *Times* and *Amounts* variable observations between the sample firms (columns (5)–(6) of Table 5, Panel B).

Table 4
Robustness tests.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables	Ab_CDD	Ab_CDD	CDD	CDD	CDD	CDD	CDD	CDD	CDD	CDD
Times	0.108***		0.140***		0.091***		0.071***		0.212***	
	(3.34)		(2.91)		(4.17)		(4.08)		(3.19)	
Amounts		0.007***		55.476*		0.009***		0.007***		0.012***
		(3.23)		(1.91)		(4.71)		(4.61)		(3.17)
Constant	−1.092***	−1.091***	−1.009***	−1.018***	−1.024***	−1.022***	−1.022***	−1.020***	−0.612**	−0.612**
	(−4.95)	(−4.95)	(−3.74)	(−3.77)	(−3.80)	(−3.79)	(−3.79)	(−3.78)	(−2.36)	(−2.36)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	34,750	34,750	34,750	34,750	34,750	34,750	34,750	34,750	34,750	34,750
Adj. R^2	0.021	0.021	0.365	0.365	0.365	0.365	0.365	0.365	0.364	0.364

Table 5
Endogeneity tests.

Panel A	(1)	(2)	(3)	(4)	(5)	(6)
PSM type	1:1	1:2	1:3	1:1	1:2	1:3
Variables	CDD	CDD	CDD	CDD	CDD	CDD
Times	0.165***	0.142***	0.140***			
	(3.51)	(3.31)	(3.42)			
Amounts				0.008***	0.009***	0.009***
				(2.59)	(3.01)	(3.10)
Constant	−2.292***	−1.977***	−1.767***	−2.649***	−2.327***	−2.234***
	(−3.46)	(−3.51)	(−3.45)	(−3.88)	(−3.92)	(−4.15)
Controls	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
N	2220	3183	4155	2216	3183	4144
Adj. R^2	0.417	0.406	0.417	0.409	0.403	0.406
Panel B	(1)	(2)	(3)	(4)	(5)	(6)
Variables	*CDD*	*CDD*	*CDD*	*CDD*	*CDD*	*CDD*
Times	0.118***		0.016**		0.043	
	(3.35)		(2.07)		(1.32)	
Amounts		0.007***		0.001**		0.001
		(3.10)		(1.99)		(0.58)
Tone	−0.045***	−0.045***				
	(−20.33)	(−20.33)				
Readability	1.286***	1.286***				
	(19.58)	(19.58)				
Spvol	0.001	0.001				
	(0.77)	(0.78)				
Constant	−1.944***	−1.943***	−3.879***	−3.880***	−1.009***	−1.008***
	(−7.66)	(−7.66)	(−9.01)	(−9.01)	(−3.74)	(−3.73)
Controls	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Firm FE	NO	NO	YES	YES	NO	NO
N	33,775	33,775	34,241	34,241	34,750	34,750
Adj. R^2	0.425	0.425	0.706	0.706	0.365	0.365

3.4. Potential channel tests

Empirical results indicate that insider opportunistic trading plays an important role in explaining firms' digitalization disclosure behavior in the earlier period. This section examines how CDD helps insider trading, providing evidence for why insiders disclose more digitalization information before reducing their holdings. Previous studies have found that high investor sentiment can raise stock prices (Firth et al., 2015). Therefore, we infer that triggering investor optimism sentiment is one of the motivations for insiders to disclose digitalization information before reducing their holdings. To verify this, we use the natural logarithm of one plus the number of positive posts on the Oriental Wealth website's stock bar within 10, 20, and 30 days after the annual report publication to measure investor sentiment (*Sentiment*) and conduct channel tests, as shown in Table 6. The coefficients of *CDD* are all significantly positive, indicating CDD can boost investor sentiment.[1]

3.5. Moderating effect of economic policy uncertainty

EPU increases information asymmetry between firms and external stakeholders (Nagar et al., 2019). We therefore anticipate that given the increasingly opaque corporate information environment due to EPU, insiders will have a stronger incentive to use hot information to hype the stock and sell shares at inflated prices (Firth et al., 2015). To confirm this, we use the EPU index developed by Baker et al. (2016), taking an annual arithmetic average and dividing by 100 to measure *EPU*. We then introduce *Times* × *EPU* and *Amounts* × *EPU* into model (1), as shown in Table 7. The coefficients of *Times* × *EPU* and *Amounts* × *EPU* are significantly positive, aligning with our expectations.

4. Conclusions

This paper investigates the relationship between insider trading behavior and CDD, based on the principle of economic self-interest.

[1] Oriental Wealth website's stock bar is one of the most user-accessed and influential stock online forums in the Chinese stock market.

Table 6
Potential channel tests: investor sentiment.

Variables	(1)	(2)	(3)
	Sentiment_10	Sentiment_20	Sentiment_30
CDD	0.027***	0.027***	0.028***
	(4.37)	(4.44)	(4.80)
Constant	−5.272***	−4.373***	−3.807***
	(−30.01)	(−26.02)	(−23.31)
Controls	YES	YES	YES
Year FE	YES	YES	YES
Industry FE	YES	YES	YES
N	34,750	34,750	34,750
Adj. R^2	0.310	0.352	0.404

Table 7
The moderating effect of EPU.

Variables	(1)	(2)
	CDD	CDD
Times	0.105	
	(1.57)	
Amounts		0.006
		(1.33)
Times × *EPU*	0.009*	
	(1.89)	
Amounts × *EPU*		0.001**
		(2.25)
Constant	−1.016***	−1.014***
	(−3.76)	(−3.76)
Controls	YES	YES
Year FE	YES	YES
Industry FE	YES	YES
N	34,750	34,750
Adj. R^2	0.365	0.365

The findings show that CDD in MD&A section can boost investor sentiment, leading insiders to reduce holdings when stock prices are high. Furthermore, the moderating effect suggests that EPU can worsen this phenomenon.

The proposed findings provide new insights for external investors and regulators. Investors should improve their information interpretation skills, be wary of insiders using hot topics to hype concepts during the publication of the annual report, and focus on the company's fundamental information when investing. Regulators should focus on establishing norms for textual information disclosure in annual reports, enhance the identification and supervision for distorted enterprise disclosure of textual information, thereby practically improving the capital market's information environment. In summary, both investors and regulators should exercise caution about information disclosed by firms on hot topics to guard against poor decision-making, especially during times of high EPU.

Due to limited data access, the following questions warrant further investigation: (1) How does digitalization information from other sources, such as audio conference calls and video interviews with executives, affect insider self-interested behavior? (2) The use of large language models such as Bert to classify digitalization information and study the impact of insider opportunity motivation on different types of CDD.

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CRediT authorship contribution statement

Kaolei Guan: Supervision, Funding acquisition, Conceptualization. **Mengting Fu:** Writing – original draft, Methodology, Data curation. **Haining Zhu:** Writing – review & editing, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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